Week 1: Introduction & Spatial Descriptions

EE0849 Introduction to Robotics - Detailed Notes

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1 Introduction to Robotics

1.1 History and Applications

The word “robot” was first used in Karel Čapek’s 1920 play “R.U.R.” (Rossum’s Universal Robots). The Czech word “robota” means forced labor.

Key milestones in industrial robotics:

- **1961**: Unimate, the first industrial robot, installed at GM
- **1978**: PUMA robot introduced for assembly tasks
- **Today**: Major manufacturers include FANUC, ABB, KUKA, and Yaskawa

1.2 Course Software Stack

- **Python 3** with libraries:
  - `roboticstoolbox-python`: Robot modeling and simulation
  - `spatialmath-python`: 3D transformations
  - `numpy`: Numerical computing
  - `matplotlib`: Visualization

2 Spatial Descriptions

2.1 Position

A position is a point in 3D space represented as a vector:

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

2.2 Orientation

Orientation describes how a body is rotated in space. Unlike position, which requires only 3 numbers, orientation requires a 3×3 rotation matrix.

3 Rotation Matrices

3.1 The Special Orthogonal Group $SO(3)$

A matrix R belongs to $SO(3)$ if:

1. $R^T R = I$ (orthogonality)
2. $\det(R) = 1$ (proper rotation)

3.2 Properties

- Inverse equals transpose: $R^{-1} = R^T$
- Columns are orthonormal unit vectors
- Composition: $R_{total} = R_2 R_1$

4 Elementary Rotations

4.1 Rotation about Principal Axes

4.1.1 X-axis rotation:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

4.1.2 Y-axis rotation:

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

4.1.3 Z-axis rotation:

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5 Summary

- Position is described by a 3D vector
- Orientation requires a rotation matrix
- Rotation matrices have special properties ($SO(3)$)
- Order of composition matters in rotations